

Bright learn

a Section 1 : Foundational concepts of Analytics.

01 Business Analytics

Is the use of data, stats methods, and analytical techniques to understand business performance and support better decision.

It helps organisations use information to solve business problems, identify opportunities, and improve their results

Realworld example.

A retail company can analyse its sales data to determine which products sell the most and use this information to decide how much stock to order.

Source = IBM (n.d) what is business analytics? IBM,

02 Data Analytics

The process of examining, cleaning, transforming, and interpreting data to find useful patterns and information.

Real world Example

A hospital could analyse patient data to identify patterns in appointment cancellations and use the findings to improve its scheduling process.

Source: IBM (n.d) what is data analytics? IBM.

03 Business Intelligence

Refers to the technologies, processes, and tools businesses use to turn data into information that can support decision making.

Real world example

A Supermarket manager could use a Power BI dashboard to monitor daily sales, revenue, stock levels & performance of different stores

Source: Microsoft (n.d) what is business intelligence? Microsoft

④ Data driven decision making

Using relevant and reliable data as an important basis for making business decisions rather than relying only on assumptions or personal opinions.

Real world example

An online clothing business might analyse website traffic and sales data before deciding which products to promote during a marketing campaign.

Source: Provost, F and Fawcett, T. (2013) Data Science for Business
O'Reilly media.

⑤ Big data

Extremely large and complex datasets that can be difficult to process using traditional data processing methods.

Real world example

A social media company can collect millions of posts, search comments, videos, and interactions every day and analyse this information to understand user behaviour.

Source: IBM (n.d) What is big data? IBM.

⑥ Descriptive Analytics

Focuses on understanding what has already happened by examining historical data. 1.

Real world example

A clothing store could create a monthly report showing that sales increased by 15% in December compared with November.

Source: Gartner (n.d) Descriptive analytics, Gartner Glossary.

⑦ Diagnostic Analytics

Investigates why something happened by examining data more deeply and looking for relationships, patterns, or possible causes. It often involves comparing different groups or sections within the data to find explanations.

for a result

Real world example

If a company's sales decreased in March, an analyst could investigate whether the decline was caused by fewer customers, higher prices, stock shortages, or a reduction in advertising.

Source: Gartner (n.d) Diagnostic analytics. Gartner & WSOJ

⑧ Predictive Analytics

Uses historical data, statistical techniques, and sometimes machine learning to estimate what could happen in the future.

Real world example

A bank would use information about previous customers to predict which applicants may have a higher risk of failing to repay a loan.

Source: IBM (n.d) What is predictive analytics? IBM.

⑨ Prescriptive Analytics

Goes beyond predicting what might happen by helping determine what action could be taken in response. It can use data, predictions, rules, and optimisation techniques to compare possible actions and support decision-making.

Real-world example

An airline could use demand forecasts and other information to help determine how many seats to make available at different prices. Source: IBM (n.d) What is prescriptive analytics? IBM.

⑩ Data literacy

Is the ability to understand, work with, analyse, and communicate using data. A data-literate person should be able to interpret information correctly, recognise misleading conclusions, and use data appropriately when making decisions.

Real-world example

A marketing employee who understands data literacy can look at a campaign dashboard and correctly interpret whether an increase in website visits actually resulted in more customers.

Source: Microsoft (n.d) what is data literacy? Microsoft.

Section 2: Data, Measurement & Reporting.

(11) Structured data:

Information organised in consistent format, usually in rows and columns, making it relatively easy for computers to store, search, and analyse.

Examples include customer databases, sales spreadsheets & transaction records.

Real world example

A supermarket may store customer transactions in a database containing columns for customer ID, product, quantity, price and purchase date.

Source: IBM (n.d) what is structured data? IBM.

(12) Unstructured data:

Is information that does not follow a fixed table like structure. It can include text, photographs, videos, audio recordings, social media posts, and emails which often requires specialised methods to analyse.

Real world example:

A company could analyse customer reviews on social media to understand how customers feel about its products.

Source: IBM (n.d) what is unstructured data? IBM.

⑬ Data warehouse

Is a central system that stores large amounts of data collected from different sources so that it can be analysed and reported on. Businesses use data warehouse to bring information together and make it easier to examine historical trends & performance.

Real-world example

A retail company could combine sales information from its physical stores, website, and mobile application into a data warehouse for management reporting.

Source: IBM (n.d.) What is a data warehouse? IBM.

⑭ Data-mining

Is the process of examining large datasets to discover useful patterns, relationships, trends, or unusual behaviour. Analysts can use statistical and computational techniques to find information that may not be obvious from simply looking at the raw data.

Real-world example

An online retailer could analyse customer purchasing patterns and discover that customers who buy a particular type of phone often purchase a phone case as well.

Source: IBM (n.d.) What is data mining? IBM.

⑮ Data - visualization

Is the presentation of data in a visual form, such as charts, graphs, maps, or other graphics. It makes information easier to understand and can help analysts and decision-makers identify patterns, trends, and unusual results.

Real-world example

A business could use a line chart to show how its monthly revenue has changed over the past 2 years.

Source

Microsoft (n.d) Data visualization. Microsoft

⑩ Dashboard

Is a visual display that brings important information and performance measures together in one place.

Real world example:

A sales manager could use a power BI dashboard to monitor daily sales, revenue, sales targets, and the performance of different sales teams.

Source: Microsoft (n.d) What is a dashboard? Microsoft

⑪ Metric

Is any measurable value that can be used to describe or monitor something, businesses can have many metrics but not all of them are important enough to directly influence major business decisions.

Real world example.

An online store might track the number of visitors to its website as a metric.

Source: Google Analytics (n.d) Analytics Help. Google

⑫ Key performance Indicator (KPI)

Is a specific metric that is considered important for measuring progress towards an important business objective.

Therefore, every KPI is a metric, but not every metric is a KPI

Real world example

If an online store's objective is to increase sales, its conversion rate could be a KPI because it directly helps management assess whether website visitors are becoming customers.

Source

Parameter, B. (2015) 'Key Performance Indicators: Developing, Implementing and using winning KPIs'. 3rd ed. Wiley.

(19) Benchmark

Is a reference point used to compare performance and determine whether a result is relatively high, low or on target.

Real world example

A call centre may compare its average customer waiting time with an industry benchmark to determine whether its service is performing at an acceptable level.

Source

Investopedia. (n.d) Benchmark. Investopedia.

(20) Segmentation

Is the process of dividing a larger group of customers, products, or data into smaller groups, that share similar characteristics.

Real world example

A clothing store could divide customers into groups based on age, location, purchasing behaviour, and spending patterns then send different promotions to each group.

Source: Kotler, P. and Keller, K.L (2016) Marketing management, 15th ed. Pearson

Section 3: customer & sales KPI's

21) Conversion rate

The percentage of people who complete a desired action out of the total number of people who had the opportunity to take that action.

Real world example

If 1000 people visit an online clothing store and 50 of them make a purchase, the store has a 5% conversion rate. Source - HubSpot (n.d) customer acquisition analytics, HubSpot.

22) Customer Acquisition Cost (CAC)

Measures the average amount a business spends to gain one new customer.

Formula = $CAC = \frac{\text{Total sales + Marketing costs}}{\text{No of new customers acquired}}$

Real world example

A company spends R20000, on sales and marketing in one month and gains 100 new customers, It CAC is R200 per customer.

Source: Shopify (n.d) customer Acquisition Cost (CAC) Calculator and Reduce it. Shopify.

23) Customer lifetime value (CLV/ LTV)

Estimates the value a business can receive from a customer over the entire period that the customer does business with the company. It helps businesses understand why retaining customers can be important because a customer may make several purchases rather than buying only once.

Real world example.

If a customer spends an average of R500 per purchase, purchases four times a year, and remains a customer for three years, the business can estimate the customer's lifetime

revenue at approximately £6000 before considering costs and profit margins.

Real world example Source

Shopify (n.d) sales KPIs: 18 Examples to Track. Shopify.

(24) Churn Rate

Measures the percentage of customers who stop using a company's product or service during a specific period. It helps a business understand how many customers it is losing and can be particularly important for subscription-based businesses.

Churn rate = $\frac{\text{Customers lost during the period}}{\text{Customers at the start of the period}} \times 100$.

Real world example

A streaming service has 10,000 customers at the beginning of a month and 500 cancel their subscriptions. The monthly churn rate is 5%.

Source: Shopify (n.d) customer acquisition metrics for Growth. Shopify.

(25)

(25) Customer Retention rate

Measures the percentage of existing customers that a business keeps over a particular period.

Formula = $\text{Retention Rate} = \frac{(\text{Customers at end} - \text{new customers acquired})}{\text{Customers at start}} \times 100$

Real world example

A subscription company starts the month with 1000 customers, gains 100 new customers, and ends the month with 950 customers. Its retention rate is 85%.

Source:

HubSpot (n.d) customer retention metrics. HubSpot.

(26) Net Promoter Score (NPS)

Is a measure of customer loyalty based on how likely customers are to recommend a company, product, or service to others. Customers are generally asked to rate their likelihood of recommending the business on a scale of 0-10.

Real world Example

After using a bank's mobile application, a customer receives a survey asking, "How likely are you to recommend our bank to a friend or colleague?" The customer's response contributes to the bank's NPS.

Source: Qualtrics (n.d.) Net Promoter Score (NPS). Qualtrics

(27) Average order value

Measures the average amount of revenue generated from each customer order or transaction.

Formula: $AOV = \text{Total Revenue} \div \text{number of orders}$

Real-world example

An online store generates \$100,000 from 500 orders during a month. Its AOV is \$200 per order.

Source: Shopify (n.d.) Marketing KPIs. Shopify

(28) Sales funnel

Represents the stages a potential customer moves through before becoming a customer. Although the exact stages differ between businesses, they commonly involve becoming aware of a product, showing interest, considering the purchase, and eventually making a purchase.

Real-world example

A person sees an Instagram advertisement for a travel company, visits its website, looks at holiday packages, contacts the company for a quotation, and eventually books a trip. These steps represent movement through

the sales funnel.

Source: HubSpot (n.d.) sales funnel HubSpot.

What is the CLV: CAC ratio?

Compares how much value a business expects to receive from a customer

For example - CLV = £3000 ; CAC = £1000

∴ £3000 / £1000 = 3:1

This means that the business estimates £3 of customer lifetime value for every £1 spent acquiring the customer.

Section 4: Marketing & Campaign Analytics

29 Marketing campaign

Is a planned and coordinated set of marketing activities designed to achieve a specific objective, such as increasing sales, generating leads, or creating awareness of a product.

Real-world example

A travel company could run a campaign promoting its December holiday packages through Instagram advertisements, email marketing and its website.

Source: Kotler, P. and Keller, K. L. (2016), Marketing

30 Management. 15th ed. Pearson.

30 Post campaign Analysis.

Is the process of reviewing campaign data after a marketing campaign has finished to determine how well it performed.

Real-world example.

After running a FB campaign, a clothing company could compare the advertising cost with the sales generated and identify which advertisements produced the most customers.

Source: Google Analytics (n.d) Google Analytics documentation: Google.



③ Incremental Revenue.

The additional revenue that can be attributed to a particular activity, such as a marketing campaign, compared with what would have happened without that activity.

Real World example

A retailer normally expects ₹100,000 in sales during a particular week, but after launching a campaign its sales reach ₹130,000. If the additional ₹30,000, is shown to have been caused by the campaign, that amount represents incremental revenue.

Source:

Kohavi, R, Tang, D. and Xu, Y. (2020). Trustworthy online controlled experiments: A

③ Return on Investment (ROI)

Measure the financial return generated from an investment compared with the cost of that investment. (compared with the cost of that investment).

formula

$$ROI = (\text{Return} - \text{Investment Cost}) \div \text{Investment Cost} \times 100$$

Real-world example

A business spends ₹10,000 on a campaign and generates ₹15,000 in additional profit from the campaign. Its ROI would be 50%.

Source: Investopedia (n.d) Return on investment (ROI).
investopedia.

③ Return on Ad Spend (ROAS)

Measures the revenue generated for each unit of money spent specifically on advertising. Unlike ROI, which considers the broader return and investment costs, ROAS focuses specifically on the relationship between advertising revenue and advertising expenditure.

Formula

ROAS = Revenue Attributed to Advertising $\div$ Advertising cost.

Real-world example:

A company spends R5000 on Google Ads and generates R20000 in revenue attributed to those advertisements.

$$ROAS = R20000 \div R5000 = 4.$$

Source: Google Ads Help (n.d.) About target ROAS bidding.
Google

(34) Click Through Rate.

Measures the percentage of people who see an advertisement email or other digital content and click on it.

Real world example

An online advertisement is displayed 10000 times and receives 300 clicks. Its CTR is 3%.

Formula = Number of clicks $\div$ number of Impressions $\times 100$.

Source: Google Ads Help (n.d.) click through rate (CTR)
Google

(35) Bounce Rate

is measure used to describe website visits where visitors leave without continuing to another page or completing further engagement; depending on the analytics platform's definition:

Real world example

A company sends customers to a product landing page through an advertisement.

Source: Google Analytics Help (n.d.) Bounce rate.
Google

(36) A/B Testing

(36) A/B Testing

Is a controlled experiment where two versions of something, such as an advertisement, web page are compared to determine which version produces a better result.

Real world example.

An online store could send two versions of an email to customers. Version A has a "Shop Now" button and version B has a "view collection" button. The company can compare their click or conversion rates.

Source

Kohari, R., Tang, D. and Xu, Y. (2020) 'Trustworthy online Unbiased Experiments: A Practical Guide to A/B Testing'. Cambridge University Press

(37) Lead Generation

Is the process of attracting and identifying people who may be interested in a company's products or services and collecting information that allows the business to contact them. A lead does not necessarily become a customer, so businesses often track how many leads eventually convert into customers.

Real world example

A travel agency could advertise a holiday package and interested people to complete an online form with their name, email address, and preferred travel dates.

Source: HubSpot (n.d.) What is lead generation?

HubSpot

(38) Marketing Attribution

Is the process of analysing customer interactions with different marketing channels to determine which channels or touchpoints contributed to a conversion.

It helps businesses understand the customer journey rather than automatically giving all the credits to the final attractions.

Real-world example

A customer might first discover a travel company through Instagram, later search for the company on Google, receive an email, and finally book a holiday through the company's website. Attribution analysis can be used to examine the contribution of these different touchpoints.

Source:

Google Analytics (n.d.) Attribution. Google